Supplementary Information: Quantum-inspired catalytic QUBO search for balanced hypergraph partitioning

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The main manuscript defines the cut objectives and QUBO encoding. Here, classical baselines lead into balance correlations and coordinate barriers. The catalytic response then motivates the shared solver policy. Mechanism controls distinguish dynamical effects from classical spectral information and document failures. Computational costs delimit practical usefulness, followed by an audit of the evidence. Primary benchmarks and exploratory controls retain their specified designs and are not pooled. In cut/imbalance tables, parentheses give observed imbalance in percent. Historical cut/time tables mark the parenthesized runtime with *s*. The allowed imbalance is an upper bound, not a required imbalance.

Supplementary Note 1. Classical baselines

Multilevel search

KaHyPar optimizes the native hypergraph objective through a hierarchy of smaller problems and subsequent refinement[1]. Its preprocessing identifies communities[2] and, when the configuration's thresholds are met, applies min-hash sparsification. Coarsening contracts strongly connected vertices while tracking their combined load. This retains a coarse partitioning problem rather than replacing each net by its full clique. On the smallest hypergraph, a pool of initial partitioners finds candidate balanced assignments. Uncoarsening then lifts the selected assignment through the hierarchy and refines it at each level.

The cut-oriented configurations used here select multilevel coarsening with heavy-edge, community-aware ratings. Their initial partitioning stage uses recursive multi-level partitioning and 20 pool trials, even though the outer partitioning mode is direct. Refinement combines Fiduccia–Mattheyses (FM) gain-based vertex moves[3] with HyperFlowCutter[4]. Flow refinement optimizes a region between block pairs through max-flow/min-cut computations[5]; it can change several vertex assignments together. The classical reference thus combines multiple search scales and discrete collective moves. Its quality is not explained solely by changing an optimizer for a fixed continuous objective.

Reference protocol

Every reference receives the original hypergraph, the cut-net objective, two blocks and the requested integer capacity. Both ordinary and evolutionary runs use cut-oriented multilevel refinement. Ordinary time-limited repeated partitioning repeats independent multilevel searches.

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KaHyPar-E instead maintains and evolves a population of partitions[6]; the two modes are evaluated separately. Each result includes the seed, nominal time allowance, actual elapsed time and returned partition. The single-run enumeration controls, historical strict-balance references and matched 1%/3% runs retain their own recorded budgets; they are not presented as one uniform KaHyPar timing experiment.

Each process uses one CPU thread. Native cut and both block sizes are recomputed from its saved partition, rather than inferred from the reported objective alone. An output that violates the capacity is not a feasible comparator. Measured elapsed time is retained because the nominal limit is soft. The hardware and concurrency are specified in Supplementary Note 6. Reference partitions serve only for comparison; they never initialize FEM trajectories.

Spectral readout

A second classical control tests how much quality is already present in the incidence modes used to construct collective catalysts. Hypergraph Laplacians and spectral partitioning provide the structural background[7,8]; our readout and catalyst controls compare two uses of those same continuous directions. Each of the eight modes is stably sorted, and exactly $\lfloor n/2 \rfloor$ vertices are assigned to one block. The resulting partitions are scored on the original nets. This readout is exactly balanced and requires no FEM trajectory. It is therefore a control for classical structural information, not an additional FEM update or input partition. Its quality and charged construction time are compared with the catalytic runs in the mechanism and cost sections below.

Supplementary Note 2. Balance and dynamics

The classical references enforce capacity during discrete partitioning. For a variational binary-energy search, a quadratic balance penalty must also be evaluated while the labels remain uncertain. The same marginals can conceal different count fluctuations, and a correct binary endpoint can still obstruct coordinate descent. These are separate questions from whether the encoded cut matches the original hypergraph.

Constraint correlations

A faithful binary target still leaves a distributional question: how is balance represented before spins become discrete?

Let $p_i = \Pr(x_i = 1)$. These marginals do not specify the fluctuations of the vertex count. This distinction between marginal and joint trial distributions is central to variational inference[9]. For $N = \sum_i x_i$, $\mathrm{Var}(N) = \sum_i p_i(1-p_i) + 2\sum_{i<j} \mathrm{Cov}(x_i, x_j)$. At $p_i = 1/2$, independent vertices have variance $n/4$. A uniform fixed-cardinality distribution has the same marginals but zero count variance, with off-diagonal covariance $-1/[4(n-1)]$ for even n. Exact enumeration on example B gives expected balance costs 24 and 0, respectively, at penalty 8. Including each distribution's own entropy at temperature 1 gives free energies 23.8307 and 1.7622. Both trial families use the same cut-net QUBO and deterministic minimum-energy net auxiliaries. The correlated distribution has a higher expected cut, 8.5909 versus 8.1484, so this comparison isolates feasibility correlations rather than improved optimization.

Cardinality-conditioned or cluster trial distributions could retain these correlations, as in richer cluster trial families[10]. Their scalable evaluation and unbiased readout remain untested here. Neither this calculation nor removal of an encoding-induced barrier establishes an OGP escape.

Coordinate barriers

Quadratic constraint penalties and auxiliary elimination are standard ingredients of Ising and pseudo-Boolean formulations[11,12]. The bound below is our encoding-specific coordinate-barrier statement, checked independently on retained states.

The variance example concerns uncertain distributions. At the discrete endpoint, the same balance penalty creates a different obstruction. For an exactly balanced, auxiliary-minimized state, $\Delta_i E_{\mathrm{Q}} \geq \lambda - D_i > 0$ for every single vertex flip. This coordinate barrier follows from the encoding, even for a poor cut. The following checks test that statement on retained states; they do not test a forbidden interval of overlaps between low-cost partitions. Such a global overlap statement requires a specified low-cost ensemble and algorithm class[13,14].

The main-text coordinate-barrier lemma is checked on all 384 selected complete-QUBO states from the six-arm *finan512*/*thermal1* refinement study. Every state is exactly balanced and auxiliary-minimized, and every single-bit energy delta is nonnegative. *finan512* has $D_{\max} = 55$ and $\lambda = 56$; its smallest observed vertex-flip energy increase is 54. *thermal1* has $D_{\max} = 11$ and $\lambda = 12$; its smallest increase is 10. Already feasible raw endpoints (119 and 74, respectively) receive no vertex flips during subsequent descent. The smallest checked cuts, 438 and 2,578, still exceed strict KaHyPar cuts 169 and 291.

Holding balance aside, conditional auxiliary minimization permits some single-vertex native-cut decreases: median counts of such vertices are 18 on *finan512* and 138 on *thermal1* across the 192 states each. Those moves violate exact balance and are not applied. Thus, absence of an improving complete-QUBO one-bit move is not evidence of good partition quality. Independent finite differences check one most-favorable vertex per batch, and weighted small-instance enumeration checks every variable of every balanced assignment. This diagnostic neither updates the partitions nor uses KaHyPar states as initial conditions. It does not establish OGP or preclude collective continuous/quantum dynamics.

Supplementary Note 3. Catalytic dynamics

Temporary interactions in quantum annealing motivate changing a search path while restoring its endpoint[15,16]. Those studies do not derive the diagonal incidence couplings evaluated here or establish their behaviour on these hypergraphs.

Exact encoding alone therefore does not supply an effective descent path. The catalysts act before the discrete endpoint: temporary local and collective interactions change the variational response, and their coefficients vanish at the final target. The relevant question is whether they provide balanced directions while probabilities are still uncertain. The calculation below establishes that local mechanism. Later controls ask whether the directions improve actual partitions or merely reproduce information already available in classical modes.

Collective curvature

For a unit-norm mode $\mathbf{v}$ with $\mathbf{1}^{\mathsf{T}}\mathbf{v} = 0$, consider the quadratic catalyst

$$C_v(\mathbf{s}) = -\frac{\rho}{2}\left[(\mathbf{v}^{\mathsf{T}}\mathbf{s})^2 - \sum_i v_i^2\right], \qquad s_i = 2x_i - 1. \tag{1}$$

Writing $m_i = 2p_i - 1$, its variational probability gradient is $-2\rho[\mathbf{v}(\mathbf{v}^{\mathsf{T}}\mathbf{m}) - \mathbf{v}^{\odot 2} \odot \mathbf{m}]$.

For independent Bernoulli vertices, removing diagonal self-interactions in the collective expectation gives exactly $-4\rho(\mathbf{v}\mathbf{v}^{\mathsf{T}} - \mathrm{diag}(v_i^2))$ as its probability Hessian. The Rayleigh value along a unit mode is $-4\rho(1 - \sum_i v_i^4)$. Because $\mathbf{1}^{\mathsf{T}}\mathbf{v} = 0$, the mean-balance rank-one term contributes zero in this direction; the balance-variance diagonal term remains in the full Hessian. The argument identifies a local balanced instability, not a path through all near-optimal partitions, a quantum spectral gap or an OGP remedy. For nonnegative unit-sum mode weights, the weighted operator is the sum of these exact quadratic Hessians. Null-support directions obey the same identity. These diagonal Ising couplings would require their own connectivity and precision treatment on quantum hardware.

Variational landscape

The two-mode slice in Fig. S1 isolates the local mechanism near the uncertain state. The minimum projected curvature is 2.67 without catalysts and -1.30 with catalysts: a local bowl develops an unstable direction. This does not measure an overlap gap or a global energy barrier.

The supplementary variational slice uses $\mathbf{p} = \mathbf{1}/2 + q_1\mathbf{v}_1 + q_2\mathbf{v}_2$; zero-mean orthonormal modes keep the mean balance fixed. Vertex entropy, the full balance expectation and diagonal corrections to the collective expectation are retained. For the schematic landscapes only, a dimensionless illustration uses $V_0(x, y) = (x^2-1)^2+0.6y^2+0.25x$ and $V_g = V_0 + g[-0.55(x^2-1)^2 + 0.6x]$. The displayed paths follow gradient descent from $(1.05, 0.7)$ with step 0.025, at $g = 0$ and $g = 1$. These illustrative coordinates are not spin overlaps. Setting $g = 0$ restores V_0 exactly; this construction is not a hypergraph experiment.

These calculations establish a change in local free-energy stability. They neither demonstrate an

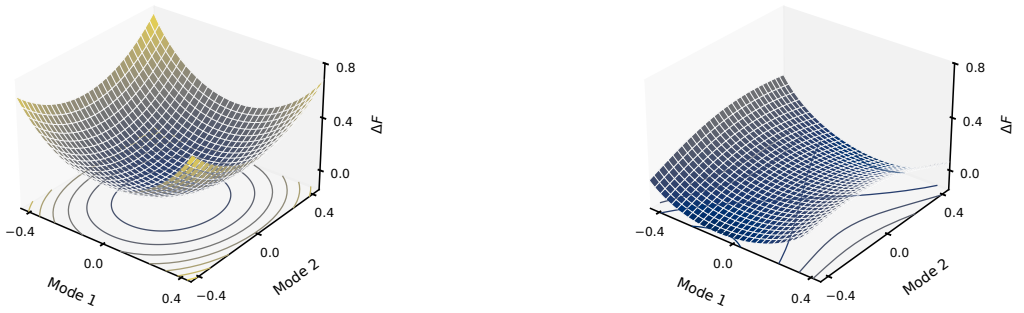


Figure S1 Variational slice; (a) Uncatalyzed and **(b)** catalyzed free energy on example B at temperature 1 and penalty 0.01, using the same two lowest nonconstant incidence modes; The catalyst has pair strength 1.3, auxiliary scale 1 and collective strength 1 with equal mode weights; Each surface shows $F(\mathbf{p}) - F(\mathbf{1}/2)$ on identical axes and color scales; every p_i lies between 0.05 and 0.95; This is a classical variational projection

overlap gap on the benchmark graphs nor prove that a catalyst removes such a gap. The shared solver and the following controls test consequences for finite-instance search.

Supplementary Note 4. Shared policy

We now turn from a local mechanism to a complete algorithm. The implicit update below evolves the original variables without a binary balance projection. The same replica allocation and schedule are then used across the primary graphs. Group-level quality and matched-capacity readout records expose what this shared policy achieves; they are not a selection of a different solver for each graph.

Implicit update

Mean-field annealing[17] and the variational FEM[18] supply the optimization framework. The implicit balance update below is evaluated for the stated trial distribution. Fields h_i give $p_i = \mathrm{sigmoid}(h_i)$ and $m_i = \tanh(h_i/2)$. The manual update separates the sparse cost gradient g_i from balance. With $d_i = \eta_a/(1 + D_i)$, set $r_i = d_i/(1 + 2\lambda_t d_i)$ and $\rho_i = r_i/(1 + r_i T_i)$. The tilted field is $\widetilde{h}_i = (h_i - r_i[g_i - \lambda_t m_i])/(1 + r_i T_i)$, clipped to $[-20, 20]$. The new field is $h_i' = \widetilde{h}_i - 2\lambda_t \rho_i \delta$, where

$$\delta = \sum_i \mathrm{sigmoid}(\widetilde{h}_i - 2\lambda_t \rho_i \delta) - b. \qquad (2)$$

The corresponding root function is strictly increasing, with derivative $1 + \sum_i 2\lambda_t \rho_i p_i'(1 - p_i') > 0$. Twenty-four bracketed Newton iterations solve this scalar equation. It is a soft-penalty update, not a projection onto a balanced binary set.

Replica allocation

The common 48-replica policy assigns 16 replicas to local catalysts, 16 to unscaled modes and 16 to normalized modes. The latter groups cycle through first mode, second mode, equal first two and normalized inverse-eigenvalue weights over all four. The inverse uses floor $\max(10^{-12}, 10^{-6} \max_\ell \mu_\ell)$ to handle null modes. These choices cross with strengths $\{0.5, 1, 2, 4\} \max(1, \bar{D})$. The local, auxiliary and per-spin profiles act simultaneously in every group. All three groups receive identical nearby initial fields. Temperature annealing follows the classical free-energy search principle[19]; the trajectories evolve independently. Exchange Monte Carlo[20] is a different mechanism and is not used by this shared policy.

Geometry groups

The original 20-instance subset uses seed 21 and one strict 48-replica batch per graph. This policy was fixed before these outcomes, after previous studies had already evaluated every graph. Within each group, the common 16 profiles and strengths are unchanged. The three groups receive matched random fields and 12,144 updates. Their

costs are charged as one batch; group quality columns below do not imply separate measured GPU times. All 960 trajectories have independently reconstructed initial fields, native cuts, full-QUBO scores, endpoint coefficients and binary states. No winning raw state needs a descent flip. Further exploratory controls do not replace this uniform primary policy.

Mode residuals, localization and component counts are in the compact study summaries. Normalization improves *IBM08* and *IBM15* but does not make the localized *ca-CondMat* directions useful. Its unscaled internal-mode inverse participation ratios are 0.99989, 0.99978 and 0.33310, compared with 0.08216, 0.05746 and 0.04777 after normalization. These remain localized directions: the catalytic curvature factor $1 - \sum_i v_i^4$ is nearly zero for the first two unscaled modes. Component contrasts and degree normalization alone do not provide a successful global direction.

Matched balance

The ten strict extension cases use the same 48 profiles, seed 21 and 12,144 updates as the original subset. Table S2 reports each group's feasible raw cut. The complete strict *scircuit* endpoint is 318 after six original-QUBO flips, compared with raw cut 373. The 3% *scircuit* solve failed both its original 24-iteration and 64-iteration residual checks before producing a complete result. Captured and uncaptured updates agree on the failing saturated field. The final retry uses 64 iterations of the same implicit equation, a central-quarter Newton acceptance bracket and the unchanged tolerance; both failed attempts (325.28 s in total) are charged separately. This numerical exception does not change the objective, initial fields or catalyst schedules. The separate 1% and 3% suites use bounded binary slack, its conditional Gibbs distribution and the full count/slack variance in the expectation. Their vertex fields and catalyst schedules match the strict suite. No strict winner is substituted into a band result, and no classical partition enters initialization.

The 1% protocol specifies 64 central-safeguarded iterations for *scircuit* before observing its new batch, and 24 iterations for the other cases. All 30 batches complete without a numerical retry. Independent reconstruction checks all 1,440 trajectories and both raw and selected encoded states. KaHyPar uses seed 0, a nominal 64 s repeated partitioning budget and one thread per process; up to four reference processes execute concurrently. The corresponding recorded process time is separate from its nominal budget.

Raw cut is the best feasible field-sign output before descent. Valid counts denote raw replicas with zero auxiliary gap and the allowed count bound, out of 48. Flips refer to the replica selected after original-QUBO descent; that replica can differ from the best feasible raw replica. All maintable solutions satisfy their configured bound. These counts and cut differences distinguish feasibility from objective quality.

Supplementary Note 5. Mechanism controls

The shared benchmark measures the resulting solver, whereas these controls ask which ingredients account for its behaviour. We begin with weak endpoints, test local schedules and correlations, then compare certified collective modes with classical mode rounding. Structure, timing and encoding controls finally test how specific the response is. Each study retains its own seeds, budget and endpoint: improvements on these known graphs do not replace the frozen benchmark or establish unseen transfer.

Weak endpoints

The weak-case diagnostics identify two distinct issues: probabilities can polarize around a poor cut before the final penalty ramp, and cut-inactive vertices can still affect balance. This motivates tests of the path and its correlations rather than additional endpoint descent.

The two weak cases have similar median net size, seven, but different constraint structure. *finan512* has 74,752 vertices and nets, 596,992 pins, maximum incidence degree 55 and one connected vertex component. *thermal1* has 82,654 vertices and nets, 574,458 pins and maximum degree 11. Its largest component contains 82,406 vertices; the remaining 248 vertices are incident only to singleton nets and do not affect the cut. These statistics come directly from the incidence representation, without clique expansion or a partition-dependent selection.

Table S1 Portfolio groups; Native cut (seconds);

Graph	Local	Unscaled	Normalized	Spectral (s)	KaHyPar 0% (s)
IBM01	325	496	509	476 (0.62 s)	287 (128.68 s)
IBM02	497	477	513	625 (1.88 s)	367 (128.30 s)
IBM03	1,217	1,171	1,303	1,555 (1.25 s)	1,247 (128.61 s)
IBM04	621	674	667	929 (1.26 s)	696 (128.67 s)
IBM06	1,320	1,080	1,095	1,348 (1.34 s)	1,335 (129.22 s)
IBM08	1,577	1,248	1,238	2,103 (54.31 s)	1,200 (128.37 s)
IBM09	1,504	756	805	967 (6.38 s)	703 (128.67 s)
IBM12	2,486	2,763	2,443	4,082 (22.60 s)	3,138 (128.81 s)
IBM15	5,270	5,134	4,573	6,106 (57.95 s)	4,554 (65.63 s)
IBM17	3,838	3,446	3,778	4,623 (40.36 s)	3,390 (144.28 s)
IBM18	3,522	2,197	2,396	3,518 (62.99 s)	2,772 (130.88 s)
bcsstk29	732	372	372	426 (0.84 s)	360 (129.15 s)
2cubes_sphere	5,004	4,125	4,137	4,289 (10.09 s)	6,892 (129.14 s)
G2_circuit	3,071	624	624	624 (43.55 s)	1,144 (129.09 s)
PGPgiantcompo	463	463	466	540 (1.76 s)	396 (128.57 s)
finan512	299	146	146	146 (16.05 s)	169 (128.59 s)
thermal1	5,545	255	255	272 (30.52 s)	291 (128.47 s)
airfoil_2d	413	375	376	395 (0.78 s)	372 (128.35 s)
ca-CondMat	4,473	4,522	4,522	6,035 (6.18 s)	4,202 (128.85 s)
parabolic_fem	4,490	1,026	1,026	1,026 (1467.23 s)	1,256 (133.58 s)

Table S2 Strict extension groups; Native cut;

Graph	Local	Unscaled	Normalized	Spectral	KaHyPar 0%
IBM05	1,901	1,947	1,936	2,171	1,807
IBM07	1,361	1,163	1,256	1,869	1,428
IBM10	2,694	2,619	2,070	2,569	1,626
IBM11	1,632	1,581	1,642	1,958	1,671
IBM13	2,544	2,641	2,677	2,725	1,204
IBM14	3,606	3,491	3,259	4,663	2,273
IBM16	2,907	3,368	3,845	6,068	4,044
cfd1	2,064	2,206	2,206	3,420	3,121
shallow_water2	882	768	768	768	909
scircuit	2,014	373	401	215	68

Table S4 resolves the median rows of the main table into their two seed pairs. On *finan512*, the winning raw and selected partitions coincide and receive no descent flips. Their native cuts, 1,004 and 1,064, are already fixed while the balance penalty is small. Across the diverse batches, median mean vertex variance falls to 3.57×10^{-8} at update 5,376, before the final penalty ramp. The late ramp mainly changes feasibility, rather than the large-scale cut pattern (Fig. S2). The two winning states have zero and 18 balance-ignoring native improving vertices, respectively; every complete-QUBO vertex flip still costs at least 56. These observations support early polarization and a final coordinate barrier, without establishing that polarization causes the entire quality gap.

thermal1 needs a different interpretation. None of the 16 diverse replicas in either guarded seed pair is feasible at the final checkpoint; six of 16 frozen replicas are feasible in each pair. The raw winners have cuts 3,233 and 3,444 and assign all 248 cut-inactive vertices to one block. The post-descent winners are different diverse replicas: they require 107 and 190 improving complete-QUBO flips, including 107 and 116 vertex flips, and yield cuts 2,374 and 2,430. Thus the median change

Table S3 Raw feasibility and readout

	Raw valid / 48			0%		1%		3%	
Graph	0%	1%	3%	Raw cut	Flips	Raw cut	Flips	Raw cut	Flips
IBM01	47	48	48	325	0	368	0	335	0
IBM02	44	48	48	477	0	454	0	420	0
IBM03	47	48	48	1,171	0	1,160	0	1,143	0
IBM04	46	47	47	621	0	636	0	609	0
IBM05	45	48	48	1,901	0	1,896	0	1,895	0
IBM06	47	48	48	1,080	0	1,078	0	1,080	0
IBM07	48	48	48	1,163	0	1,144	0	1,206	0
IBM08	47	47	48	1,238	0	1,181	0	1,211	0
IBM09	48	48	48	756	0	723	0	685	0
IBM10	47	48	48	2,070	0	1,732	0	1,720	0
IBM11	46	48	48	1,581	0	1,458	0	1,417	0
IBM12	45	48	47	2,443	0	2,457	0	2,492	0
IBM13	48	48	47	2,544	0	2,508	0	1,921	0
IBM14	47	48	48	3,259	0	3,088	0	3,577	0
IBM15	48	48	48	4,573	0	4,721	0	4,145	0
IBM16	47	48	48	2,907	0	3,770	0	3,544	0
IBM17	48	48	48	3,446	0	3,366	0	2,784	0
IBM18	45	46	48	2,197	0	2,231	0	2,326	0
2cubes_sphere	48	48	48	4,125	0	3,991	0	3,951	0
G2_circuit	40	48	48	624	0	624	0	624	0
PGPgiantcompo	37	48	48	463	0	467	0	457	0
airfoil_2d	48	47	48	375	0	373	0	342	0
bcsstk29	47	42	34	372	0	360	0	360	0
ca-CondMat	39	48	48	4,473	0	4,406	0	4,215	0
cfd1	48	47	48	2,064	0	2,174	0	2,250	0
finan512	45	46	48	146	0	146	0	146	0
parabolic_fem	42	48	48	1,026	0	1,026	0	1,026	0
scircuit	32	44	48	373	6	277	0	46	0
shallow_water2	38	48	48	768	0	768	0	768	0
thermal1	37	48	48	255	0	257	0	257	0

from 3,338.5 to 2,402 is a change in the winning candidate as well as its endpoint, not the improvement of one raw winner. Even the selected winners remain far above strict KaHyPar. Native-cut decreases considered in this diagnostic are never applied when they violate balance.

Local controls

Frozen and guarded portfolios

We freeze one 16-profile portfolio before evaluating six additional graphs. The four policy-selection graphs are *IBM01*, *IBM15*, *IBM06* and *bcsstk29*. Held-out graphs are *IBM02*, *IBM09*, *IBM17*, *2cubes_sphere*, *G2_circuit* and *PGPgiantcompo*. Each method receives the same nearby starts for base seeds 3 and 4, 6,000 annealing updates and 144 cold updates. The primary endpoint is raw feasible cut; disclosed one-bit QUBO descent is a separate endpoint. KaHyPar partitions are used solely as references.

Against the previous diverse portfolio, the frozen portfolio reduces the geometric mean of per-graph median raw-cut ratios to 0.9586, an improvement of 4.14%. Seven paired graph–seed comparisons improve and five regress (Fig. S3). The worst median regression is 22.97% on *IBM02*. Applying one negative-contrast/warm-hub rule to every replica performs worse: its geometric-mean ratio is 1.2056 and *G2_circuit* regresses by a factor of 3.42. Thus, even a locally interpretable rule can fail to transfer. The portfolio reduces this sensitivity without eliminating it.

To retain a fallback, a guarded policy allocates one full batch to the diverse control and one to the frozen portfolio. Its comparator allocates both

Table S4 Weak-case endpoints; Cut (seconds) and selected-winner flips, by seed pair;

Graph	Seeds	Guard raw	Guard selected	Flips	Strict KaHyPar	KaHyPar 3%
finan512	5/6	1,004 (24.89 s)	1,004 (26.17 s)	0	169 (128.59 s)	146 (64.52 s)
	7/8	1,064 (24.88 s)	1,064 (26.17 s)	0	169 (128.59 s)	146 (64.52 s)
thermal1	5/6	3,233 (24.20 s)	2,374 (25.93 s)	107	291 (128.47 s)	239 (66.02 s)
	7/8	3,444 (24.29 s)	2,430 (26.20 s)	190	291 (128.47 s)	239 (66.02 s)

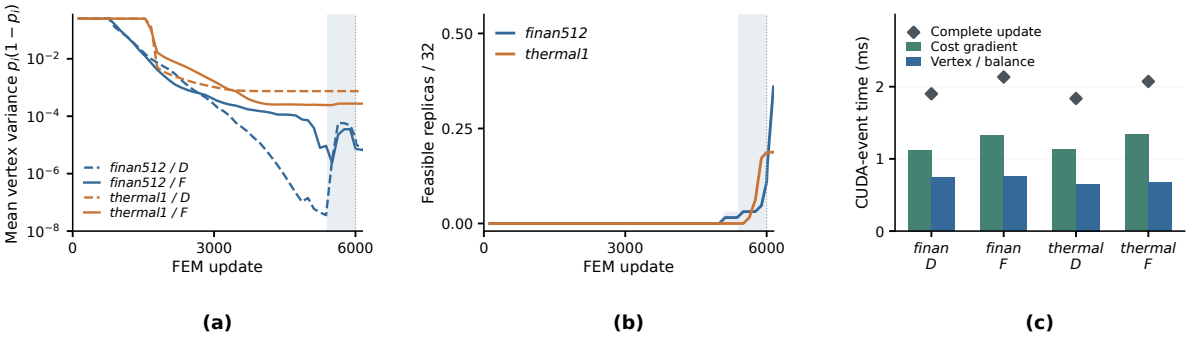


Figure S2 Failure profiles; (a) Mean vertex variance per replica, summarized by a replica median and then the average of two base seeds; Dashed curves are diverse (D); solid curves are frozen (F); The shaded interval is the final balance penalty ramp; **(b)** Feasible fraction of the 32 guarded replicas; lines average seed pairs and bands show their range, not confidence intervals; Feasibility requires exact balance and zero auxiliary gap; **(c)** Separate A100 CUDA-event timings of the cost-gradient stage, vertex/balance update and complete update at the cold endpoint; Component timings replay fixed inputs; complete timings restore and repeat a 32-update window; They are separate measurements, not a stacked decomposition of one event

Table S5 Guarded endpoints; Median cut (seconds); two seed pairs;

	Guarded FEM, 32 trajectories		Restart, 32		
Graph	Raw	With descent	With descent	Strict KaHyPar	KaHyPar 3%
IBM04	680 (12.44 s)	680 (12.85 s)	700 (12.15 s)	696 (128.67 s)	582 (65.10 s)
IBM12	2,879 (23.18 s)	2,879 (24.27 s)	3,150 (23.10 s)	3,138 (128.81 s)	1,936 (66.35 s)
finan512	1,034 (24.89 s)	1,034 (26.17 s)	908 (24.86 s)	169 (128.59 s)	146 (64.52 s)
thermal1	3,338.5 (24.24 s)	2,402 (26.06 s)	2,362 (25.02 s)	291 (128.47 s)	239 (66.02 s)

Table S6 Local portfolios; Median cut (FEM seconds); seeds 3/4;

Graph	Diverse FEM	Frozen FEM	Strict KaHyPar	KaHyPar 3%
IBM01	426 (4.50 s)	285.5 (4.96 s)	287 (128.68 s)	202 (64.67 s)
IBM15	5,480 (20.70 s)	4,959.5 (22.83 s)	4,554 (65.63 s)	2,741 (69.89 s)
IBM06	1,690 (6.15 s)	1,363 (6.92 s)	1,335 (129.22 s)	988 (66.49 s)
bcsstk29	528 (6.80 s)	1,026 (7.81 s)	360 (129.15 s)	360 (64.41 s)
IBM02	418 (5.10 s)	514 (5.76 s)	367 (128.30 s)	337 (64.44 s)
IBM09	1,787 (8.80 s)	1,442 (9.84 s)	703 (128.67 s)	620 (64.66 s)
IBM17	4,381 (23.82 s)	3,807 (26.15 s)	3,390 (144.28 s)	2,310 (65.19 s)
2cubes_sphere	8,817 (19.48 s)	9,325 (22.28 s)	6,892 (129.14 s)	3,884 (155.80 s)
G2_circuit	3,482.5 (17.69 s)	3,165.5 (19.37 s)	1,144 (129.09 s)	624 (67.27 s)
PGPgiantcompo	517 (4.21 s)	484 (4.64 s)	396 (128.57 s)	350 (64.40 s)

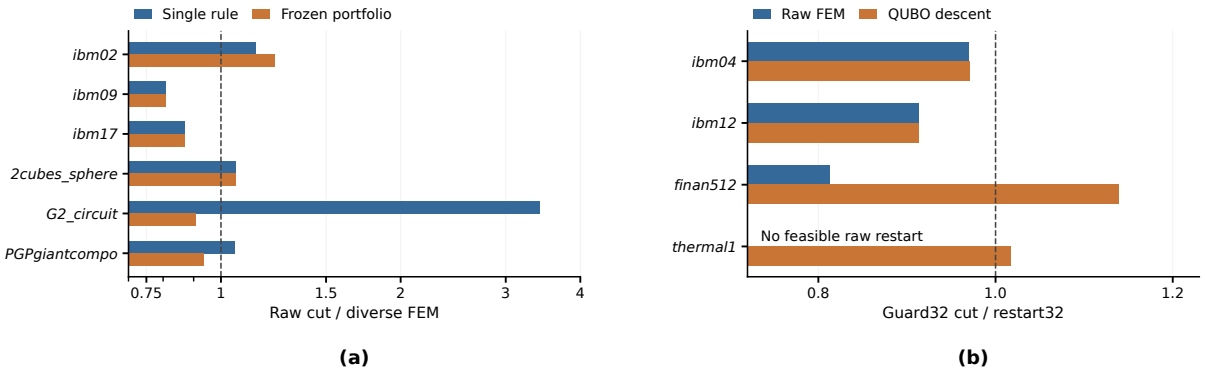


Figure S3 Portfolio transfer; (a) median raw-cut ratios of the single-rule and frozen portfolios against the diverse control on six held-out graphs; **(b)** guarded 32-trajectory ratios against two ordinary restarts on four further graphs; Values below one improve the corresponding QUBO comparator; Raw and post-descent endpoints are separated; *thermal1* has no finite raw ratio because the comparator returns no feasible raw output; Medians use two seeds or seed pairs; the plot does not imply confidence intervals

batches to independent diverse restarts. Each policy spends 32 trajectories. The common measured control component is charged to both; only four further graphs count as validation of this follow-up.

Guarding improves all six paired comparisons with finite raw outputs and returns feasible states in the two *thermal1* cases where ordinary restarts fail. We retain those failures, so the specified four-graph raw geometric-mean ratio is undefined. After one-bit QUBO descent, however, the guarded policy is 0.70% worse overall, with a 13.88% worst median regression on *finan512* (Table S7). Both variants remain far behind KaHyPar on *finan512* and *thermal1*. Keeping a complete reference batch protects its own result but does not establish superiority to an equally large restart budget.

Annealing and probe controls

thermal1 separates numerical accuracy from discrete feasibility. The implicit-root residual is of order 10^{-12}, while thresholded ordinary replicas have balance residuals spanning approximately -124 to $+123$. Summed vertex Bernoulli variance has median about 62, compared with 19–26 for the frozen catalyst batches. These marginals are more polarized, but neither precise root solving nor lower variance guarantees a high-quality balanced partition. One trajectory also exhausts its 50,000-flip descent budget with a nonzero auxiliary gap. Its state remains in the record and is excluded from feasible quality scores.

We test six fixed refinement arms on *IBM02*, *IBM09*, *finan512* and *thermal1* using two fresh seeds, giving 768 further trajectories. These previously tested weak graphs are exploratory cases, not another held-out set. Auxiliary-only minimization at readout leaves the thresholded vertices unchanged. Lowering the terminal temperature from 0.02 to 0.0002 does not consistently improve the final cut. Allocating 6,144 updates to the cold phase returns the same best raw and post-descent cuts as the shorter cold phase in all eight paired cases. A longer anneal is more useful on *finan512*: its median complete-QUBO cut falls from 1,032.5 to 537, with individual cuts 438 and 636, at nearly twice the FEM time. It remains well above strict KaHyPar's 169.

The collective extension adds a quadratic reward for alignment with graph-filtered smooth modes, while retaining random initial fields. It regresses on all four post-descent comparisons, including *IBM09* cuts of 10,642 on both seeds against 1,442 for the control. On *thermal1*, the four probes have participation fractions of only 0.10–0.16%: their amplitude is concentrated on a small vertex set. Between 83% and 87% of squared amplitude lies on 248 singleton-only vertices, which have no response in the pair Laplacian. This is localization of a classical filter, not quantum many-body localization. A separate

Table S7 Guarded controls; Median cut (seconds); two seed pairs;

Graph	Two restarts	Guarded catalysts	Strict KaHyPar	KaHyPar 3%
IBM04	700 (12.15 s)	680 (12.85 s)	696 (128.67 s)	582 (65.10 s)
IBM12	3,150 (23.10 s)	2,879 (24.27 s)	3,138 (128.81 s)	1,936 (66.35 s)
finan512	908 (24.86 s)	1,034 (26.17 s)	169 (128.59 s)	146 (64.52 s)
thermal1	2,362 (25.02 s)	2,402 (26.06 s)	291 (128.47 s)	239 (66.02 s)

follow-up excludes those vertices from the filter while retaining every original QUBO spin. Their mode amplitude becomes zero, but the complete cut is 3,718 on both seeds, worse than the original collective result of 3,404. The support defect is real, yet correcting it alone does not yield a useful search path (Fig. S4).

Let $\mathbf{B}$ be the net-by-vertex incidence matrix and $a_e = w_e / \max(1, \lfloor r_e^2/4 \rfloor)$. The implicit clique Laplacian is $\mathbf{L} = \mathrm{diag}(\mathbf{B}^\mathsf{T}(\mathbf{a} \odot \mathbf{r})) - \mathbf{B}^\mathsf{T} \mathrm{diag}(\mathbf{a})\mathbf{B}$. We diffuse four fixed random probes in the zero-mean subspace using $\mathbf{V} \leftarrow \mathbf{V} - \mathbf{PLV}/M$, where $\mathbf{P}$ subtracts column means and $M = 2 \max_i \sum_{e \ni i} a_e(r_e - 1)$ (bounded below by 2). The filter uses 512 updates and QR orthogonalization every 32. These are smooth probes, not certified lowest eigenvectors.

Strengths are the mean weighted incidence degree, bounded below by one, times 0.5, 1, 2 or 4, crossed with four probes. The original catalyst envelope restores $C_v = 0$ before the cold phase. Preprocessing time is charged separately. The low-rank representation makes classical evaluation inexpensive but implies dense long-range Ising couplings; quantum hardware connectivity and precision costs are not evaluated here.

The refinement campaign also compares the original auxiliary readout with auxiliary-only minimization conditional on the thresholded vertices. Only auxiliaries change; binary balance is never repaired. Colder tests end at temperature 0.0002. A 12,144-update control allocates 12,000 updates to annealing and 144 to cooling; its matched-budget alternative allocates 6,000 to annealing and 6,144 to the cold phase. These exploratory arms cannot be called held-out validation after their graphs influenced design.

The support follow-up uses only *thermal1* and the same base seeds. It masks vertices with zero $\sum_{e \ni i} a_e(r_e - 1)$ in the probe filter and subtracts means on the remaining support. All original vertices remain in the variational states, QUBO, balance equation and scoring. The filter budget, catalyst strengths, profiles and 6,144-update search budget remain unchanged. It adds 32 trajectories and is a targeted exploratory follow-up, not validation on new graphs.

Pair correlations

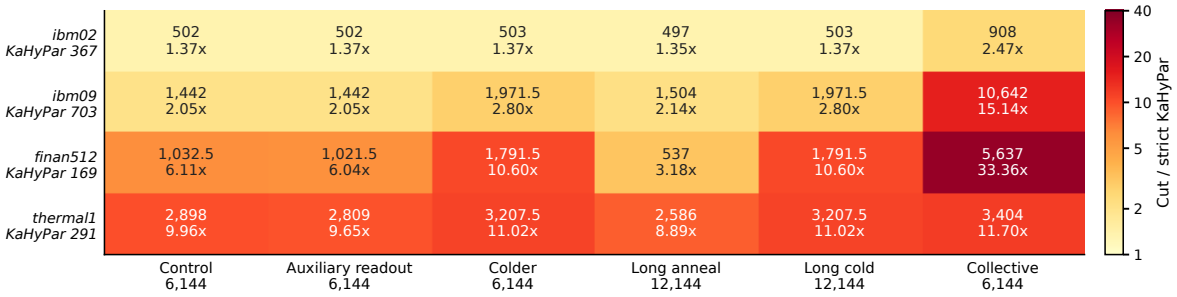
The balance-variance calculation at the beginning suggests retaining vertex covariance. The following control changes the trial distribution while keeping the original binary problem; its cost and failures delimit what this additional variational freedom achieves.

To test covariance beyond independent vertex marginals, we replace vertex factors by joint four-state pair distributions. Block-factor trial distributions are a standard extension of the independent approximation[9,10]; the following controls test that extension for this cut-net energy. All four assignments remain available, and an odd vertex is a singleton. Auxiliary Gibbs blocks remain factorized from vertex factors. The full binary QUBO and its original variables are retained; no logical-group restriction, incumbent partition or binary exchange move is used. Random factors are predetermined per replica; local factors greedily pair available vertices sharing a net, with random fallback. All methods retain the frozen catalytic profiles, nearby original fields, seeds 11/12 and 6,144 updates.

For independent factors g with occupancy S_g, the exact balance expectation is

$$\mathbb{E}[(S-b)^2] = (\textstyle\sum_g \mathbb{E}S_g - b)^2 + \sum_g \mathrm{Var}(S_g). \quad (3)$$

Within-pair covariance also corrects the temporary quadratic pair cost. At nonuniform temperatures, joint entropy uses $T_i H_-(p_i) + T_j H_-(p_j) + \min(T_i, T_j)\, \mathrm{I}(x_i; x_j)$. Here H_- is binary negative entropy and I mutual information. This equals the old per-spin entropy at independence and joint negative entropy at uniform temperature.



(a)

Figure S4 Refinement controls; (a) Cells show median post-descent cuts over two fresh seeds and their ratios to the strict KaHyPar reference printed beside each graph; Column labels give updates per trajectory; The colder and long-cold arms retain the same best outputs despite different cold budgets; The longer anneal improves *finan512* but remains above the reference; All six arms are exploratory seen-graph tests, use 16 trajectories per batch and retain the original QUBO; the five refinement arms disclose auxiliary-only minimization at readout; Complete cut/time tables, including strict and 3% references, are retained in the experiment report

Table S8 Refinement controls; Median cut (seconds); short-budget block above, long-budget block below;

Graph	Control	Aux readout	Colder	Collective	Strict KaHyPar	KaHyPar 3%
ibm02	502 (5.86 s)	502 (5.86 s)	503 (5.83 s)	908 (5.91 s)	367 (128.30 s)	337 (64.44 s)
ibm09	1,442 (10.28 s)	1,442 (10.28 s)	1,971.5 (10.29 s)	10,642 (10.61 s)	703 (128.67 s)	620 (64.66 s)
finan512	1,032.5 (13.79 s)	1,021.5 (13.77 s)	1,791.5 (13.81 s)	5,637 (14.20 s)	169 (128.59 s)	146 (64.52 s)
thermal1	2,898 (13.73 s)	2,809 (13.63 s)	3,207.5 (13.69 s)	3,404 (14.10 s)	291 (128.47 s)	239 (66.02 s)

Graph	Long anneal	Long cold	Strict KaHyPar	KaHyPar 3%
ibm02	497 (11.41 s)	503 (11.28 s)	367 (128.30 s)	337 (64.44 s)
ibm09	1,504 (19.93 s)	1,971.5 (19.90 s)	703 (128.67 s)	620 (64.66 s)
finan512	537 (26.69 s)	1,791.5 (26.83 s)	169 (128.59 s)	146 (64.52 s)
thermal1	2,586 (26.22 s)	3,207.5 (26.44 s)	291 (128.47 s)	239 (66.02 s)

The monotone implicit root uses 40 iterations versus 24 in the independent control. The singleton limit agrees with the original equations when clipping is inactive; weighted QUBO expectations, covariance corrections and entropy derivatives agree with independent enumeration and automatic derivatives. Joint log weights span at most 80, whereas the previous tilted vertex fields clip at 20. Factor preconditioning and root iterations also differ, so this is a complete-solver comparison, not an isolated covariance ablation.

The completed 384-trajectory comparison is in Table S9. Local pairs reduce selected *IBM12* median cut by 4.48%, *finan512* by 0.84% and *IBM04* by 0.07%, but worsen *thermal1* by 26.84%. FEM time increases 2.89–4.28-fold. Random pairs remove all infeasible *finan512* readouts in 32 trajectories, versus 13 for the independent control, but have worse selected cut. Pair covariance can reduce balance variance without giving better partitions. These are exploratory seen-graph tests, not further held-out evidence for the frozen primary policy.

Factor refresh

Different fixed factors still have zero covariance between their occupancies. A globally balanced distribution with uncertain factor totals would need cross-factor anticorrelation. We therefore test predetermined random rematching at updates 4,096, 5,120 and 5,632 on the two weak graphs, with the same seeds and initial random factors. It preserves vertex marginals to FP64 precision while discarding old pair covariance. This can raise expected balance/catalyst energy; the jumps are recorded. It is a variational restart operation, not an energy-preserving physical catalyst. Partner choices never use an incumbent partition.

Table S9 Pair-correlation controls; Median cut (seconds); raw and selected endpoints;

Conditional-auxiliary readout; FEM seconds					
Graph	Independent	Random pairs	Local pairs	Strict KaHyPar	KaHyPar 3%
finan512	1,213.5 (13.22 s)	1,177 (41.83 s)	1,065.5 (42.27 s)	169 (128.59 s)	146 (64.52 s)
thermal1	3,358.5 (12.72 s)	3,330.5 (54.48 s)	3,575.5 (54.43 s)	291 (128.47 s)	239 (66.02 s)
IBM04	680 (6.44 s)	680 (18.65 s)	679.5 (18.57 s)	696 (128.67 s)	582 (65.10 s)
IBM12	3,043.5 (12.07 s)	3,058.5 (46.00 s)	2,907 (45.90 s)	3,138 (128.81 s)	1,936 (66.35 s)
Complete-QUBO descent; FEM + descent seconds					
Graph	Independent	Random pairs	Local pairs	Strict KaHyPar	KaHyPar 3%
finan512	1,074.5 (13.96 s)	1,177 (42.51 s)	1,065.5 (42.95 s)	169 (128.59 s)	146 (64.52 s)
thermal1	2,434.5 (13.47 s)	2,853 (55.22 s)	3,088 (55.18 s)	291 (128.47 s)	239 (66.02 s)
IBM04	680 (6.66 s)	680 (18.87 s)	679.5 (18.79 s)	696 (128.67 s)	582 (65.10 s)
IBM12	3,043.5 (12.65 s)	3,058.5 (46.60 s)	2,907 (46.56 s)	3,138 (128.81 s)	1,936 (66.35 s)

These 64 further trajectories give essentially unchanged selected cuts relative to fixed random pairs (Table S10). *thermal1*'s feasible pre-descent median improves from 3,330.5 to 2,856.5, but selected median cut changes from 2,853 to 2,856.5. *finan512* changes from 1,177 to 1,176.5. Neither weak case is solved by this reconfiguration. All 448 new initial and endpoint states are independently scored. The follow-up audit verifies all 128 unchanged-prefix checkpoints through update 4,096, marginal preservation at each refresh and final seed-generated pairings. The refresh preserves the specified marginals under GPU execution. An independent calculation also checks the entropy contribution under nonuniform temperatures. These controls are evaluated separately from the timing measurements in Supplementary Note 6.

Collective controls

The weak-probe results motivate a sharper test: certified low-frequency directions rather than a fixed number of diffusion steps. The experiments below distinguish their dynamical effect from simply rounding the same classical modes.

The accurately converged spectral construction differs from the earlier 512-step diffusion probes. The latter were not certified low-frequency modes and gave poor partitions. Here the active Laplacian is resolved by Lanczos iteration[21]; the main text defines the resulting QUBO terms. The resolved-mode control contains 18 populations, 288 trajectories, base seeds 15/16 and 12,144 updates. Every arm retains the same local portfolio, original random fields, penalty schedule, stopping rule and auxiliary-only conditional readout. KaHyPar and binary spectral partitions are never supplied to the solver.

A signed/hold control reverses the local pair sign early, restores it later and anneals from 8 to 0.35 over the first 70%, then to 0.02. The matched long control instead uses the ordinary exponential schedule. Mode-only catalysis gives cut 146 on *finan512* for both seeds. On *thermal1* it gives no exactly balanced raw output; the selected cut is 276. Adding the fixed random null-support coupling gives raw cut 255 for both seeds, with zero winning descent flips. The signed/hold combination instead gives 264. The 248 zero-response *thermal1* vertices remain in the full QUBO throughout; they are neither deleted nor balanced afterward. Table S11 retains the failed raw endpoint and all control cuts.

Transfer

The unchanged single-mode/null policy next evaluates *IBM02* and *G2_circuit* at seeds 17/18, adding 128 trajectories. These graphs were previously seen under other policies, but were not used to choose the new spectral strengths or schedules. The test therefore measures transfer of this rule, not fully unseen-graph validation. *G2_circuit* reaches 624 for both seeds, compared with the matched long control's post-descent median 2,687.5. *IBM02* reaches 482 versus 497. Both graphs have no inactive support, so their null terms are identically zero.

A further 32-trajectory, seen-*IBM02* follow-up replaces individual modes by weighted subspaces: equal weights on the first two, first three or all four modes, and normalized inverse-eigenvalue weights on four. It reaches 457 for both seeds, still above

Table S10 Factor-refresh controls; Median selected cut (seconds); seeds 11/12;

Graph	Independent	Fixed random	Refreshed	Strict KaHyPar	KaHyPar 3%
finan512	1,074.5 (13.96 s)	1,177 (42.51 s)	1,176.5 (42.85 s)	169 (128.59 s)	146 (64.52 s)
thermal1	2,434.5 (13.47 s)	2,853 (55.22 s)	2,856.5 (55.18 s)	291 (128.47 s)	239 (66.02 s)

Table S11 Collective ablations; Median cut (seconds); a dash denotes no feasible raw output;

Graph	Arm	Raw	Selected	Strict KaHyPar	KaHyPar 3%
finan512	Long control	610 (26.01 s)	610 (26.71 s)	169 (128.59 s)	146 (64.52 s)
finan512	Signed/hold	621.5 (26.01 s)	528 (26.70 s)	169 (128.59 s)	146 (64.52 s)
finan512	Modes	146 (26.54 s)	146 (27.20 s)	169 (128.59 s)	146 (64.52 s)
finan512	Modes/hold/null	146 (26.58 s)	146 (27.28 s)	169 (128.59 s)	146 (64.52 s)
thermal1	Long control	5,546.5 (25.15 s)	2,301.5 (25.95 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Signed/hold	3,248.5 (24.86 s)	2,000.5 (25.64 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes	–	276 (26.82 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes/null	255 (26.76 s)	255 (27.52 s)	291 (128.47 s)	239 (66.02 s)
thermal1	Modes/hold/null	264 (26.86 s)	264 (27.63 s)	291 (128.47 s)	239 (66.02 s)

strict KaHyPar's 367 and 364. The fourth *IBM02* mode has inverse participation 0.4665; its localization contrasts with the extended modes on the two matrix-derived failures. These observations motivate mode diversity but do not prove that localization causes the remaining gap.

Mode diversity and classical references

The mixed portfolio combines first-mode, second-mode, equal-first-two and inverse-eigenvalue subspace couplings, crossed with the four common strengths. Its 128 trajectories use exactly the same rule on all four graphs and the same seeds as their corresponding controls. The policy was fixed before mixed outcomes, after all four single-mode outcomes were known; all mixed evaluations are therefore seen-graph tests. It retains cuts 146, 255 and 624 on *finan512*, *thermal1* and *G2_circuit*, but gives 477 on *IBM02*. The separate subspace result 457 is not substituted for 477 in the shared-policy comparison. All eight raw winning states have exactly balanced vertices, zero auxiliary gap and zero descent flips.

A classical reference stably sorts each of the same four modes and selects exactly $\lfloor n/2 \rfloor$ vertices. The best native cut is reported after independently scoring all four bisections. It matches *finan512* and *G2_circuit* and is faster there. *thermal1* improves from its 272 to the catalyst's 255; *IBM02* improves from 775 to 477, but still loses to KaHyPar. This control was added after catalyst outcomes and is not evidence of a prespecified held-out comparison. Its binary outputs are never passed to FEM.

Additional strict KaHyPar-E runs use seed 1 with 256 s nominal on *finan512*/*thermal1* and 128 s on *IBM02*/*G2_circuit*. The seed/budget combination is distinct from the primary seed-0 reference and is retained with its own actual runtime. Longer nominal time does not imply a lower cut across different seeds. No minimum cut is paired with another run's shorter time.

Structure and timing

Useful directions still leave the allocation of catalytic strength and time unresolved. These controls alter structural weighting, coupling envelopes and local-temperature envelopes while retaining matched starts. Their regressions show why separately helpful interventions cannot be assumed to combine into a uniformly better method.

We test five matched 16-replica paths on six seen graphs, with seed 21, 12,000 annealing and 144 cold updates. All 480 trajectories run on the A100; a completed 80-trajectory *IBM02* run on the 4090 is retained only as a diagnostic and excluded from this comparison. Independent audits reconstruct every initial field, score every raw and selected binary state, verify zero final catalyst coefficients and check one-bit minima. No winning raw state requires a descent flip.

The control uses unscaled incidence modes with equal first-two, first-three, all-four or inverse-eigenvalue weights. An orthogonal Hadamard rotation changes the orientations selected by replicas while preserving the four-dimensional span. A coherence control weights auxiliary polarization

Table S12 Collective transfer controls; Median cut (seconds); seeds 17/18;

Graph	Arm	Raw	Selected	Strict KaHyPar	KaHyPar 3%
IBM02	Long control	497 (11.33 s)	497 (11.48 s)	367 (128.30 s)	337 (64.44 s)
IBM02	Modes/null	482 (11.52 s)	482 (11.67 s)	367 (128.30 s)	337 (64.44 s)
G2_circuit	Long control	8,457.5 (38.08 s)	2,687.5 (39.35 s)	1,144 (129.09 s)	624 (67.27 s)
G2_circuit	Modes/null	624 (39.86 s)	624 (41.05 s)	1,144 (129.09 s)	624 (67.27 s)
IBM02	Subspaces	457 (11.71 s)	457 (11.85 s)	367 (128.30 s)	337 (64.44 s)

Table S13 Collective and classical references; Native cut (charged seconds); both strict KaHyPar seeds are retained;

Graph	Mixed raw	Spectral	KaHyPar strict 0	KaHyPar strict 1	KaHyPar 3%
finan512	146 (37.67 s)	146 (6.26 s)	169 (128.59 s)	184 (257.22 s)	146 (64.52 s)
thermal1	255 (41.53 s)	272 (9.94 s)	291 (128.47 s)	303 (257.13 s)	239 (66.02 s)
IBM02	477 (16.42 s)	775 (1.61 s)	367 (128.30 s)	364 (128.40 s)	337 (64.44 s)
G2_circuit	624 (62.08 s)	624 (15.20 s)	1,144 (129.09 s)	1,274 (129.40 s)	624 (67.27 s)

using

$$\phi_e = \frac{\|\sum_{i\in e} \mathbf{v}_i\|^2 - \sum_{i\in e} \|\mathbf{v}_i\|^2}{(r_e - 1)\sum_{i\in e} \|\mathbf{v}_i\|^2}, \qquad (4)$$

where $\mathbf{v}_i$ is the row of four modes, and zero denominators give zero. Multipliers $\exp(c\phi_e)$ use $c = 0.5, 1, 1.5, 2$ across replica groups, with the total auxiliary coupling mass conserved. A heating control adds the centered, standardized and clipped log leverage $\ln \sum_\ell v_{i\ell}^2$ to the existing temporary spin-temperature field. The combined control uses all three modifications. Coherence and leverage depend on the subspace rather than its orthogonal basis. Every modification vanishes at the final original-QUBO endpoint.

Coherence weighting improves *IBM02* from 457 to 443 and *IBM08* from 1,248 to 1,235. Leverage heating improves *IBM09* from 739 to 711 and *ca-CondMat* from 4,543 to 4,529. Other cases regress or tie. The combined control improves none of the six controls and ties only *bcsstk29*. These outcomes show that independently useful structural signals need not combine constructively. They are exploratory comparisons on known inputs; they do not replace the shared 48-replica policy or establish unseen transfer.

Each arm charges the shared mode and graph costs, its own setup, the whole 16-replica FEM batch and all descents. Arms execute sequentially in one process and reuse compilation caches, so later setup times are smaller. The reported totals retain these measured components; the table does not establish an algorithmic speedup among arms.

Channel timing

Inhomogeneous quantum driving can alter an annealing path[22]. Our per-spin temperatures are classical variational controls, so retiming them does not simulate that transverse-field driver. The timing controls separate coupling changes from local-temperature changes. Four arms use identical fields, mode arrays and 48-replica budgets on each of two previously seen graphs. They retain 12,000 exponential temperature updates, 144 cold updates and the original capacity-penalty schedule. One arm remaps pair, auxiliary, collective and null coupling envelopes; another remaps only the per-spin temperature envelope; their combination matches the earlier joint intervention. Global temperature is unchanged. All catalysts vanish at the endpoint. Independent checks verify all 384 raw and selected trajectories (768 score checks), integer capacity and original-QUBO one-bit minima. Independent checks confirm that each arm changes only its specified schedule channels.

On *IBM13*, coupling-only retiming changes cut 2,544 to 2,514, whereas temperature-only retiming reaches 2,233 and joint retiming 2,211. The larger change therefore comes from local-temperature timing in this comparison. On *rajat26*, all retimed arms regress: the control is 170, coupling-only and joint controls give 202, and temperature-only gives 193. Matched KaHyPar cuts are 1,204 and 32. These controls identify a graph-dependent response, rather than a transferable replacement for the primary policy. Rajat26 was previously withheld but is now a seen instance; it is not a new hold-out test in this factorial comparison.

Table S14 Structural controls; Native cut (charged seconds); Bold marks row minima for cut and time;

Graph	Control	Rotation	Coherence	Heating	Combined	KaHyPar 0%
IBM02	457 (18.31 s)	476 (14.53 s)	443 (**14.37** s)	560 (14.50 s)	495 (14.51 s)	**367** (128.30 s)
IBM08	1,248 (78.37 s)	1,413 (**74.05** s)	1,235 (74.12 s)	1,432 (74.09 s)	1,500 (74.17 s)	**1,200** (128.37 s)
IBM09	739 (35.18 s)	1,719 (27.20 s)	758 (**27.10** s)	711 (27.21 s)	833 (27.26 s)	**703** (128.67 s)
bcsstk29	402 (26.02 s)	402 (17.55 s)	402 (17.49 s)	402 (**17.45** s)	402 (17.55 s)	**360** (129.15 s)
PGPgiantcompo	465 (20.20 s)	470 (12.01 s)	466 (12.08 s)	492 (**11.96** s)	497 (11.98 s)	**396** (128.57 s)
ca-CondMat	4,543 (23.74 s)	4,580 (19.26 s)	4,543 (19.08 s)	4,529 (19.25 s)	4,562 (**19.04** s)	**4,202** (128.85 s)

The certified GPU-constructed modes on *rajat26* are shared across all arms. Censored primary CPU solves are reported separately; timing here charges the certified mode cost, setup, search and readout. These sums do not replace the primary table or establish equal-hardware speedups.

Replica timing mixtures

We next test whether distributing a fixed QUBO search budget across several catalytic schedules improves the returned cut. One fixed mixture assigns four replicas within each 16-replica geometry group to each of the original, coupling-retimed, local-temperature-retimed and jointly retimed paths. A Latin-square allocation balances the four mode/strength choices without using graph identity. Global temperature and penalty are unchanged. All 144 new QUBO trajectories pass independent scoring; the mixture adds no replicas and no best-of-old/new selection.

The shared-to-mixed cuts are 146 to 146 on *finan512*, 2,444 to 1,941 on *IBM13*, and 209 to 228 on *rajat26*. The mixed policy improves one input and regresses on another. It spreads a finite budget across more paths, reducing coverage of individual geometry/timing combinations. This is a plausible tradeoff, rather than a demonstrated cause of the regression. Results are not inserted into the frozen primary table and do not establish unseen transfer.

The following QUBO control tables are separate from the 30-input primary tables. Cuts include measured imbalance (%) in parentheses. Bold marks the lowest cut or time in each row. Charged sums include certified historical mode costs for catalytic arms, fresh setup, search and all-replica descent; off arms do not charge unused modes. GPU-constructed modes on *rajat26* are diagnostic. Mode loading, serialization and censored primary CPU construction are excluded from these sums and retained separately. Different hardware and budgets preclude equal-hardware speedup claims.

Penalty control

A fixed hardening schedule can be replaced by a classical feedback rule. Every 128 updates, let f_j be the fraction of replicas whose thresholded original-vertex labels lie within the requested integer count bounds. The provisional penalty follows

$$\bar{\lambda}_{j+1} = \mathrm{clip}\left(\bar{\lambda}_j \exp[0.5(0.75 - f_j)], 0.01, \lambda_*\right), \tag{5}$$

where the last two arguments are the clipping bounds and $\lambda_* = D_{\max} + 1$. The effective penalty is the larger of this feedback value and a prescribed floor. The floor rises exponentially from 0.01 to λ_* between annealing fractions 0.85 and 0.95; the remaining updates and 144 cold updates restore λ_*. Thus the terminal binary QUBO and capacity bounds are unchanged. Feedback uses the current classical trajectory readouts. Because the floor also differs from the baseline's final-10% ramp, these controls do not isolate feedback from hardening time. A matched-floor control is needed before assigning a causal gain to feedback alone.

All arms retain 48 replicas, seed 21, the same initial fields, sparse operators and single-bit original-energy readout. Six policies on *IBM01* use complete 3,000-update anneals. Selected policies are then checked with 12,000 updates: feedback at 1%, and earlier hardening with warmer cooling at 3%. Warmer cooling reaches $T = 0.4$ at fraction 0.65 before cooling to 0.02. The earlier fixed ramp spans 0.65–0.9. On *IBM13*, the baseline and feedback use 3,000 updates at both capacities. These shorter runs complete their own annealing schedules; they are not prefixes or early checkpoints of a 12,000-update trajectory. All 20 QUBO executions and 1,920 raw/selected states

Table S15 Timing attribution; Native cuts; all partitions have 0% observed imbalance; bold marks the better matched cut;

Graph	Original	Coupling only	Temperature only	Both	KaHyPar
IBM13	2,544	2,514	2,233	2,211	**1,204**
rajat26	170	202	193	202	**32**

Table S16 Timing-control cost; Charged component sums in seconds; bold marks the lower time relative to KaHyPar;

Graph	Original	Coupling only	Temperature only	Both	KaHyPar
IBM13	**83.76**	**84.50**	**84.49**	**84.24**	130.33
rajat26	**113.76**	**113.51**	**113.63**	**113.54**	128.54

Table S17 QUBO cut and measured imbalance;

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	516 (0.000)	**146** (0.000)	**146** (0.000)	169 (0.000)
IBM13	5,150 (0.000)	2,444 (0.000)	1,941 (0.000)	**1,204** (0.000)
rajat26	722 (0.000)	209 (0.000)	228 (0.000)	**32** (0.000)

pass independent native-cut, capacity, energy and reported one-bit-minimum checks.

Table S19 reports the selected comparisons, including regressions. On *IBM01*, feedback at 1% worsens the cut from 311 to 392. Earlier hardening with warmer cooling at 3% does not retain the gain suggested by the shorter schedule. Feedback on *IBM13* yields no substantial improvement. All remain above the corresponding KaHyPar reference cuts. Feedback and simple combinations therefore do not yet define a robust shared improvement.

GPU trials run sequentially. Charged times in Table S20 include both cached mode-construction charges, fresh setup/capture, search, checkpoints and all 48 readouts. Baselines run first, so cache effects and single-trial variation preclude attributing small time differences to the control itself. KaHyPar runs on one CPU thread with a nominal 64 s limit; its actual time is retained. These measurements are not pooled with the primary tables or claimed as equal-hardware speedups.

Taken together, the controls support graph-dependent catalytic responses and expose failures of local schedules, covariance changes and mode choices. They do not identify a universally improving combination. That is why the primary table retains one shared policy and all its outcomes, rather than selecting the best exploratory arm per graph.

Supplementary Note 6. Computational cost

A useful catalytic direction can be expensive to obtain or evolve. We therefore separate partition quality from computational cost. The reported time includes mode construction, setup, search and readout. Larger-input and kernel studies then locate the bottlenecks, while a dense-action control tests the computational value of retaining structure. Diagnostic component timings are not substituted for benchmark elapsed time.

Primary timing

The complete pipeline time charges CPU construction of both mode geometries, input/graph setup, compilation and capture, FEM checkpoints and all 48 single-bit descents. The two CPU jobs can overlap, so their sum is a conservative charged cost rather than a parallel critical-path measurement. Reference times retain their original CPU execution and balance setting. The primary KaHyPar references use one thread of an Intel Core i7-12700. The kernel-fusion, penalty-feedback and single-precision controls use an Intel Xeon E5-2620 v3 CPU for their references. Primary reference runs allow at most two concurrent CPU processes. GPU search uses an NVIDIA

Table S18 QUBO charged time (s);

Graph	Off	Shared catalyst	Mixed catalyst	KaHyPar
finan512	**67.82**	104.78	106.21	128.59
IBM13	**80.51**	128.38	130.39	130.33
rajat26	**46.10**	89.80	91.48	128.54

Table S19 Penalty-control cut and imbalance (%);

Graph	Updates	Allowed (%)	Encoding	Baseline	Control	KaHyPar
IBM01	12,000	1	QUBO	311 (0.988)	392 (0.988)	**211** (0.988)
IBM01	12,000	3	QUBO	314 (2.651)	328 (2.996)	**202** (2.760)
IBM13	3,000	1	QUBO	1,142 (1.000)	1,146 (1.000)	**849** (0.240)
IBM13	3,000	3	QUBO	1,123 (1.128)	1,141 (1.121)	**848** (0.404)

Table S20 Penalty-control time (s);

Graph	Updates	Allowed (%)	Encoding	Baseline	Control	KaHyPar
IBM01	12,000	1	QUBO	28.36	**27.46**	64.47
IBM01	12,000	3	QUBO	28.51	**27.78**	64.30
IBM13	3,000	1	QUBO	41.59	**41.05**	66.59
IBM13	3,000	3	QUBO	**41.28**	41.36	64.87

A100 with 40 GB memory. These hardware differences are retained when interpreting the measured times.

Scaling

Six further inputs retain the frozen 16-profile policy and its 6,000 exponential annealing plus 144 cold updates. The two base seeds, 13 and 14, are fixed before outcomes. All 192 initial, raw and selected states pass independent NumPy reconstruction and scoring. The inputs cover three additional netlists, a mesh, a collaboration network and a larger finite-element instance. They extend timing coverage without changing parameters after observing quality.

Table S24 reports incidence sizes and the full time decomposition. Numerical setup is reused between runs: the first *IBM03* runner takes 8.46 s, while its second-seed runner takes 0.39 s. The table therefore retains both setup values rather than presenting a reused setup value as universal startup cost. The total is the median of per-seed graph construction, runner setup, whole-batch FEM and all-replica descent times. It excludes process startup, serialization and independent auditing. Reported GPU memory includes the arrays retained for execution and is not total device or process memory. The main table keeps raw and descent endpoints separate; for *parabolic_fem* they are 4,552 and 4,548.

Each input receives the same primary strict KaHyPar-E reference, seed 0 and a 128 s nominal limit, plus ordinary 3% KaHyPar with a 64 s nominal limit. Ordinary strict runs with 128 s nominal are also retained (Table S25). Evolutionary KaHyPar is not better on every stochastic run: *IBM18* yields 2,772 versus 2,766 for the ordinary strict run. The predefined primary comparator is retained for every graph; neither baseline nor FEM endpoint is chosen by whichever makes a quality comparison look favourable. All 18 new reference partitions are independently rescored from retained files.

GPU profile

Four additional instrumented 16-replica trajectories use the original *finan512* and *thermal1* graphs, base seed 5 and unchanged diverse/frozen profiles. They are excluded from quality and runtime medians. Five CUDA-event repeats each replay 64 fixed-input component graphs; complete update timings repeat the same restored 32-update window. Profiles are taken at updates 1,024, 4,096, 5,888 and 6,144, separately from a 16-update

Table S21 Pipeline timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Total
IBM01	0.61	0.78	10.62	13.05	0.29	25.35
IBM02	1.86	0.55	10.75	17.82	0.43	31.42
IBM03	1.23	0.71	10.89	19.84	0.51	33.19
IBM04	1.23	0.68	10.82	21.57	0.61	34.91
IBM06	1.31	0.71	10.80	23.79	0.70	37.31
IBM08	54.26	0.91	11.26	38.17	1.08	105.68
IBM09	6.33	0.98	10.72	40.34	1.37	59.74
IBM12	22.54	1.29	10.90	54.92	1.58	91.24
IBM15	57.80	3.16	11.43	132.51	4.07	208.97
IBM17	40.18	3.09	11.74	155.57	4.94	215.52
IBM18	62.79	2.72	12.00	169.73	5.53	252.76
bcsstk29	0.83	0.66	10.23	20.06	0.69	32.47
2cubes_sphere	9.99	2.18	11.44	105.52	3.80	132.94
G2_circuit	43.46	2.00	11.47	103.10	3.48	163.52
PGPgiantcompo	1.75	0.46	10.37	12.04	0.25	24.87
finan512	16.01	1.23	11.12	56.25	1.84	86.46
thermal1	30.45	1.57	11.55	61.08	2.27	106.93
airfoil_2d	0.76	0.81	10.31	16.01	0.44	28.33
ca-CondMat	6.17	0.64	11.35	22.23	0.61	40.99
parabolic_fem	1466.81	6.91	13.10	467.52	17.31	1971.66

Table S22 Strict extension timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Total
IBM05	0.84	0.87	10.63	22.77	0.64	35.75
IBM07	4.02	0.88	11.02	32.86	1.00	49.79
IBM10	18.50	1.14	11.21	51.79	1.55	84.20
IBM11	11.24	1.31	10.70	53.63	1.76	78.65
IBM13	24.93	1.41	10.97	63.78	2.18	103.27
IBM14	80.66	2.04	11.45	114.51	3.52	212.18
IBM16	126.68	2.73	11.41	152.33	4.61	297.77
cfd1	8.34	1.57	11.07	70.01	3.31	94.29
shallow_water2	10.43	1.19	11.14	49.92	1.73	74.41
scircuit	3205.45	2.56	11.89	127.70	4.58	3352.19

kernel trace. FP64, 24 implicit-root iterations, grouped CSR, compiled dense kernels and single-update CUDA Graph replay match the solver configuration. Numerical evaluation uses PyTorch[23] and CUDA Graph capture/replay[24]; sparse matrix products use cuSPARSE[25]. Two uninstrumented 16-replica *finan512* repeats match all eight profiling checkpoints in native cut, encoded cost and balance; their 32 final states are independently scored. Each component timing starts from the same numerical state. The instrumented trajectories differ slightly from the benchmark at the final *finan512* checkpoint (up to one cut net and six vertices of balance residual across replicas); the two measurements are therefore reported separately.

At update 6,144, the cost-gradient stage accounts for 59–65% of a complete update, and the vertex/balance stage for 32–39% (Table S27). A kernel trace attributes 30–34% of device duration to CSR matrix multiplication and 21–28% to the repeated implicit-root reduction kernel. The cost-gradient stage also includes auxiliary updates, structural weighting and gather/scatter operations, so its time should not all be attributed to sparse products. The fixed capture continues to compute structural gradients even when their

Table S23 3% pipeline timing; Seconds;

Graph	Modes	Graph	Runner	FEM	Descent	Failed	Total
IBM01	0.61	0.75	6.29	13.24	1.09	0.00	21.98
IBM02	1.86	0.65	6.75	18.18	2.02	0.00	29.46
IBM03	1.23	0.63	5.96	20.07	1.90	0.00	29.79
IBM04	1.23	0.90	6.71	21.81	2.35	0.00	33.00
IBM05	0.84	0.65	6.30	23.19	2.46	0.00	33.43
IBM06	1.31	0.71	6.25	24.31	2.74	0.00	35.32
IBM07	4.02	0.86	7.28	33.64	4.08	0.00	49.88
IBM08	54.26	0.88	6.96	39.05	4.55	0.00	105.71
IBM09	6.33	0.98	6.48	40.94	4.54	0.00	59.28
IBM10	18.50	1.31	6.83	52.70	5.98	0.00	85.31
IBM11	11.24	1.19	6.41	54.42	7.56	0.00	80.82
IBM12	22.54	1.18	6.61	55.45	5.95	0.00	91.72
IBM13	24.93	1.65	7.06	65.75	7.17	0.00	106.56
IBM14	80.66	2.02	6.57	117.32	13.44	0.00	220.02
IBM15	57.80	2.43	6.78	133.91	15.89	0.00	216.81
IBM16	126.68	3.14	6.74	154.13	16.63	0.00	307.32
IBM17	40.18	2.62	6.70	157.22	17.03	0.00	223.75
IBM18	62.79	2.69	6.96	171.77	19.64	0.00	263.85
2cubes_sphere	9.99	1.94	7.05	105.88	9.93	0.00	134.80
G2_circuit	43.46	2.77	7.51	106.69	12.98	0.00	173.42
PGPgiantcompo	1.75	0.70	5.93	12.33	0.90	0.00	21.61
airfoil_2d	0.76	0.64	5.82	16.25	1.34	0.00	24.81
bcsstk29	0.83	0.66	5.83	20.23	1.91	0.00	29.45
ca-CondMat	6.17	0.62	6.27	22.38	2.08	0.00	37.52
cfd1	8.34	1.65	6.61	70.75	7.78	0.00	95.12
finan512	16.01	1.49	6.63	57.45	6.72	0.00	88.30
parabolic_fem	1466.81	7.09	7.95	471.56	54.50	0.00	2007.91
scircuit	3205.45	2.67	5.70	190.23	15.25	325.28	3744.58
shallow_water2	10.43	1.47	6.26	50.11	7.04	0.00	75.30
thermal1	30.45	1.34	6.54	61.33	7.36	0.00	107.03

Table S24 Input sizes and timing; Memory in MiB; elapsed time in seconds; Runner setup is given per seed;

Graph	Vertices	Nets	Pins	Binary variables	Memory
IBM03	23,136	27,401	93,573	77,938	76.6
IBM08	51,309	50,513	204,890	152,335	134.7
IBM18	210,613	201,920	819,697	614,453	531.5
airfoil_2d	14,214	14,214	259,688	42,642	58.8
ca-CondMat	23,133	23,133	186,936	69,399	74.1
parabolic_fem	525,825	525,825	3,674,625	1,577,475	1,400.8

	Graph	Runner setup				
Graph	setup	Seed 13	Seed 14	FEM	Descent	Total
IBM03	0.52	8.46	0.39	6.09	0.19	11.22
IBM08	0.73	4.72	0.41	9.40	0.40	13.10
IBM18	2.64	5.28	0.80	28.36	1.85	35.88
airfoil_2d	0.25	4.16	0.37	6.09	0.15	8.76
ca-CondMat	0.31	4.02	0.56	6.64	0.45	9.68
parabolic_fem	7.20	5.57	1.27	79.32	6.15	96.09

Table S25 KaHyPar references; Native cut (actual CPU seconds);

Graph	Ordinary strict, 128 s	Evolutionary strict, 128 s	Ordinary 3%, 64 s
IBM03	1,438 (128.58 s)	1,247 (128.61 s)	957 (64.51 s)
IBM08	1,346 (128.61 s)	1,200 (128.37 s)	1,140 (64.52 s)
IBM18	2,766 (129.48 s)	2,772 (130.88 s)	1,915 (67.20 s)
airfoil_2d	396 (128.53 s)	372 (128.35 s)	324 (64.23 s)
ca-CondMat	4,402 (129.09 s)	4,202 (128.85 s)	3,855 (66.07 s)
parabolic_fem	1,414 (130.79 s)	1,256 (133.58 s)	1,026 (80.87 s)

Table S26 Runtime instances; Median cut (seconds); seeds 13/14;

Additional graph	Frozen raw	With descent	Strict KaHyPar	KaHyPar 3%
IBM03	1,242 (6.09 s)	1,242 (6.28 s)	1,247 (128.61 s)	957 (64.51 s)
IBM08	1,624 (9.40 s)	1,624 (9.80 s)	1,200 (128.37 s)	1,140 (64.52 s)
IBM18	3,468 (28.36 s)	3,468 (30.21 s)	2,772 (130.88 s)	1,915 (67.20 s)
airfoil_2d	397 (6.09 s)	397 (6.25 s)	372 (128.35 s)	324 (64.23 s)
ca-CondMat	4,545 (6.64 s)	4,545 (7.08 s)	4,202 (128.85 s)	3,855 (66.07 s)
parabolic_fem	4,552 (79.32 s)	4,548 (85.48 s)	1,256 (133.58 s)	1,026 (80.87 s)

cold-phase coefficients are zero. Removing such work is a possible acceleration target, requiring equivalence checks before changing the benchmark implementation. These profiles locate computational cost; they do not explain an overlap gap or improve a partition by themselves.

GPU kernel fusion

A separate numerical control combines fused dense GPU kernels, sparse CSR products, 32-update GPU execution graphs and 64 safeguarded count-root iterations. Mode construction uses GPU LOBPCG with component deflation and QR/Rayleigh–Ritz polishing[26,27]. It retains 48 replicas, FP64, seed 21, 12,000 annealing and 144 cold updates. The strict-capacity QUBO comparison includes 30 inputs, of which 29 succeed. On *scircuit*, the unscaled mode residual 5.38×10^{-4} exceeds the fixed 10^{-5} certificate; its 11.99 s failed construction is included as a failure. Original-operator residual and orthogonality limits remain fixed.

Across the successful runs, QUBO charges 186.15 s mode construction, 33.58 s setup, 3,252.37 s search/checkpoints and 82.38 s readout, totaling 3,554.48 s. Search therefore accounts for 91.5% of the charged total. Each independent solve charges its mode construction when the same modes are reused. The sum of these charges therefore differs from the elapsed time needed to execute the full set of experiments. Table S28 separates search from the complete pipeline on five inputs and gives the corresponding CPU reference times. Different hardware and mode-construction methods limit direct speedup comparisons with the primary results.

A separate instrumented *IBM01* graph-replay window contains 25,824 captured kernel calls. Device kernels occupy 164.44 ms of its 189.76 ms wall interval (86.7%). Count-root kernels account for 35.1% of kernel time, cuSPARSE products for 24.6% and other kernels for 40.2%. Instrumentation and this subrange's boundaries prevent substituting these figures for complete search time. Complete large-input kernel occupancy and bandwidth counters are not measured. The trace points to the root reductions and sparse actions as acceleration targets; it does not show that either alone explains cut quality or an overlap gap.

Single precision

To test sensitivity to arithmetic precision, we repeat the *IBM01* search with single-precision vertex fields, interaction products and count updates. TensorFloat-32 matrix multiplication is disabled. The primary benchmark and preceding kernel-fusion control use double precision and are reported separately. Exact integer cut and capacity checks are unchanged.

Table S27 GPU timing; CUDA-event milliseconds per update; memory in MiB;

Graph	Profile	Complete	Cost gradient	Vertex/balance	CSR forward	Memory
finan512	Diverse	1.901	1.123	0.738	0.139	209.6
	Frozen	2.134	1.323	0.755	0.140	235.3
thermal1	Diverse	1.836	1.135	0.652	0.128	236.6
	Frozen	2.072	1.342	0.672	0.129	257.2

Table S28 Fused-kernel time (s);

Graph	Modes	Search	Total	KaHyPar
IBM01	6.33	19.73	**27.01**	128.82
IBM18	9.13	254.25	272.33	**131.20**
finan512	6.45	83.54	**93.37**	129.01
thermal1	9.20	97.00	**109.79**	129.81
parabolic_fem	40.47	714.09	778.44	**131.67**

Count roots, auxiliary responses, sparse products and collective fields are checked against independent double-precision calculations. The separate mode-construction control uses FP32 sparse Lanczos actions[21], followed by CPU-double projection, QR and small Rayleigh–Ritz analysis. Where needed, a CPU-double LOBPCG correction[26] is charged to preprocessing; GPU actions remain FP32. On *IBM01*, both mode sets pass the unchanged 10^{-5} residual and 10^{-8} orthogonality/balance limits. Their charged construction times total 3.96 s, including 0.39 s CPU correction. This single case does not establish mode-construction scaling.

Continuous root diagnostics use $\max(10^{-4}, 16\epsilon_{\mathrm{mach}}n)$ in FP32, versus $\max(10^{-7}, 10^{-11}n)$ in FP64; this accounts for reduction roundoff rather than changing discrete feasibility. For the three complete *IBM01* QUBO precision controls, the largest GPU root residual is 1.22×10^{-4}. All 288 raw/selected binary states pass independent cut, capacity, encoded-energy and reported one-bit-minimum checks. A separate three-run QUBO check also audits 288 states with newly constructed FP32 modes. It gives the same winning cuts in this case.

Tables S29 and S30 reuse the same certified CPU mode arrays across precisions, casting them only for the solver, and retain all measured mode charges. The original random draws, 48-replica policy, 12,000 annealing and 144 cold updates are unchanged. Search takes 14.32–14.60 s in FP32 versus 19.73–20.09 s in FP64, a 1.37–1.38 ratio on this seen input. Complete charged time falls from 27.01–27.82 s to 21.66–22.39 s. The baseline and single-precision measurements were made in separate runs, so this comparison does not establish a reproducible speedup across inputs. Cut quality is sensitive to precision: strict QUBO worsens from 325 to 438, whereas 3% QUBO improves from 314 to 284. Thus the faster arithmetic does not preserve the previous winning trajectory or establish a shared quality improvement. Broader paired measurements are required to assess the generality of these changes.

Mode construction

The large *parabolic_fem* cost has two measured sources. A repeat of the original single-thread mode construction needs 15,375 unscaled and 13,974 normalized sparse products. Those products consume 341.31 and 486.48 s within total construction times of 1249.03 and 1293.90 s. The remaining time is outside the instrumented products. These diagnostic repeats run separately from the primary measurements and do not replace the original 1466.81 s charged preprocessing cost.

On A100, CUDA-event profiling of the 48-replica *parabolic_fem* update measures 38.66 ms at the initial fields and 38.52 ms at a constructed saturated field. Sparse/catalyst gradients take approximately 27.3 ms; the 24-iteration balance solve takes 9.5–9.8 ms. The measured update cost is consistent with the 467.52 s primary FEM loop. Four balance iterations take about 3.3 ms and closely reproduce the 24-iteration result at these two diagnostic fields. This is not a convergence

Table S29 Precision-control cut and imbalance (%);

Graph	Allowed (%)	Encoding	FP64	FP32	KaHyPar
IBM01	0	QUBO	325 (0.000)	438 (0.000)	**318** (0.000)
IBM01	1	QUBO	311 (0.988)	368 (0.988)	**211** (0.988)
IBM01	3	QUBO	314 (2.651)	284 (2.102)	**202** (2.760)

Table S30 Precision-control time (s);

Graph	Allowed (%)	Encoding	FP64	FP32	KaHyPar
IBM01	0	QUBO	27.01	**21.66**	128.82
IBM01	1	QUBO	27.76	**22.32**	64.47
IBM01	3	QUBO	27.82	**22.39**	64.30

guarantee for every intermediate state, so all primary quality runs retain 24 iterations. The saturated field is constructed from a saved binary state; it is not a saved continuous checkpoint.

For *scircuit*, the original unscaled shifted-Lanczos construction was stopped after 2776.60 s without a completed mode set. A component-deflated, Jacobi-preconditioned LOBPCG block solve[26] on the same incidence operator took 163.77 s on CPU; a further 11.37 s of CuPy refinement on A100 certified all four directions. CuPy[27] supplies the GPU array and sparse-linear-algebra backend; the original operator residual, rather than solver termination alone, is the certificate. Independent full-operator residuals are below 10^{-10}; balance and orthogonality errors are below 3×10^{-11} and 6×10^{-14}. This alternative was specified before evaluating partition quality on this instance. It supplies continuous coefficient modes, never a binary partition. The charged time includes the unsuccessful CPU attempt, CPU warm start, GPU refinement and separate normalized construction. The warm refinement time is not a standalone GPU construction time. A separate CuPy solve from random continuous vectors takes 19.08 s and independently reaches residuals below 10^{-10}. It is an acceleration diagnostic and does not replace the primary charged timing.

A separate full-precision incidence-product comparison on *parabolic_fem* compares sparse-product implementations. For 48 replicas, Torch CSR takes 6.53 ms forward and 6.67 ms transposed; CuPy CSR takes 1.82 ms in each direction. A CUDA warp kernel takes 1.73 ms including dense-input transposition, or 1.14 ms with prepacked input. All outputs agree with the Torch products to within 3.6×10^{-15}. The 16-replica products show smaller gains. These measurements isolate sparse products; the custom path is not substituted into the primary quality runs, and no full-solver speedup is inferred.

Collective timing

The maximum absolute spectral residuals are below 4×10^{-8}. *finan512* and *thermal1* modes have inverse participation of order 2×10^{-5}, whereas the fourth *IBM02* mode is strongly localized. Eigenmode construction is single-thread CPU work on the i7-12700, charged in every spectral pipeline total even when the saved modes are reused. The full pipeline includes graph construction, compilation/capture, FEM checkpoints and all-replica descent, excluding process startup, serialization and independent auditing. Each seed's setup time is retained to expose compilation/cache effects. The mixed solver uses FP64 manual gradients, grouped CSR operations, compiled dense kernels and single-step CUDA Graph replay on A100. Its rank-four products add collective couplings without storing dense QUBO matrices. This storage saving does not remove quantum embedding cost.

Structured and dense actions

The capacity coupling is all-to-all but has rank one plus a diagonal; its field is computed by a vertex sum. Net pair couplings use exact incidence products, and collective couplings retain their low-rank factors. No explicit dense constraint matrix is needed. A bounded A100 pilot on *IBM01* uses 12,752 vertices, 48 replicas, FP64 and the same fields for structured and explicit dense actions.

Table S31 Mixed-portfolio timing; Seconds; runner setup is given per seed;

Graph	Modes	Graph	Runner 1	Runner 2	FEM	Descent	Total
finan512	6.20	1.41	4.64	0.15	26.98	0.68	37.67
thermal1	9.87	1.31	4.95	0.17	27.08	0.70	41.53
IBM02	1.60	0.81	4.24	0.09	11.70	0.15	16.42
G2_circuit	15.11	2.08	4.62	0.24	41.13	1.33	62.08

CUDA Graph32 times are medians of five samples of 64 actions. All action differences are below 7×10^{-13}. Dense construction is excluded from action time and recorded separately. This pilot is not a whole-FEM timing comparison.

These measurements distinguish inexpensive structured evaluation from the cost of constructing a useful direction. They also separate isolated kernel gains from whole-solver gains. A lower action time does not by itself improve a cut or certify a quantum implementation.

Supplementary Note 7. Validation and scope

Independent validation separates correctness of the objective and readout from claims about search quality and generality. We reconstruct initial fields, rescore complete binary states and verify integer capacities. The netlists and matrix-derived inputs come from the KaHyPar benchmark collection[28] and the University of Florida sparse matrix collection[29]. The control studies in this Note are assessed separately from the 30-input primary benchmark.

Endpoint validation

The first frozen-policy study contains 60 populations and 960 trajectories. The guarded control adds 24 component populations and 384 trajectories; the common component is measured once and charged to each logical policy. Seed pairs are (5,6) and (7,8). All 1,344 initial states are reconstructed and raw and selected QUBO states rescored independently. All 28 retained KaHyPar partitions are checked against the same original inputs, native cuts and block sizes. Missing feasible output remains a failure. The auxiliary gap is the encoded cut term minus the native cut and must be zero for a feasible score.

Disclosed descent uses improving single-bit flips on the same complete QUBO, with a 50,000-flip cap. It is not FM, a block exchange, a cardinality repair or a KaHyPar refinement. An incomplete descent state is excluded from feasible quality and retained with its stop reason. Post-descent quality is never substituted for raw FEM in the primary endpoint. For two seeds, a median is their average and may be fractional. Geometric means use per-graph median ratios. Two seeds do not establish a population-level confidence interval or robust time-to-target estimate. Independent scoring covers 3,360 trajectories across the local-policy, readout-refinement, paired-factor, scaling and collective controls in Supplementary Notes 5 and 6, together with 50 large-instance KaHyPar reference partitions. The 96 profiling and verification trajectories are counted separately and are excluded from benchmark medians.

FEM executes in FP64 on an NVIDIA A100-PCIE-40GB, with manual gradients, grouped sparse CSR operations, compiled dense kernels and CUDA Graph replay. FEM elapsed time covers the complete batch and checkpoints, not time divided by replica count. Graph setup, runner compilation/capture and descent are retained separately. Successful measured search phases in the first two studies total 1,134.47 s, excluding process startup, auditing, output writing and interrupted setup. CPU and GPU hardware and reference concurrency are specified in Supplementary Note 6. Nominal limits are soft, so actual process times are reported. Different hardware and time budgets preclude interpreting these tables as quantum advantage, equal-hardware acceleration or equal-budget superiority.

Spectral validation

The resolved-mode, transfer, subspace and mixed controls contribute 288, 128, 32 and 128 trajectories, respectively: 576 new independently scored trajectories. Saved initial fields are reconstructed from seeds; all initial, raw and selected complete states are checked for native cut, QUBO cost, auxiliary gap and balance. Final penalty is $D_{\max} + 1$ and all catalysts are zero after 12,144 updates.

Table S32 Dense coupling cost; Milliseconds per 48-replica action;

Coupling	Structured	Dense	Dense/structured
Balance	**0.0132**	1.2822	96.79
Net pairs	**0.0997**	1.2372	12.41
Collective rank four	**0.0368**	1.5383	41.75
Net pairs + balance	**0.1217**	1.2266	10.08

Manual gradients agree with automatic derivatives, and GPU execution reproduces the direct updates for single-mode and subspace terms. All four additional KaHyPar states and all 16 classical spectral partitions are independently rescored. These checks verify the numerical construction and returned states; they do not certify global optimality.

Scope of the controls

The optimizer, temperature-schedule, degree and contrast controls test specific mechanisms on the stated inputs. They do not establish an overlap-gap remedy or general performance advantage. The main-text solver starts from random fields on the original hypergraph and uses the specified single-bit readout. Its results are assessed independently of any reference partition, with failed outputs included in the reported scope.

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